Airbnb Advertisement Strategy

Power BI dashboard: detailed analysis report

Data period: 5 May 2014 to 23 April 2015

Prepared 24 November 2025

Contents

1. Executive summary 3
- Key findings 3
- Headline recommendations 3

2. Scope, data and method 4
- 2.1 The dashboard 4
- 2.2 Data model 4
- 2.3 How the numbers in this report were obtained 4
- 2.4 Derived performance metrics 5

3. Traffic and platform mix 5

4. User demographics and spending 6
- 4.1 Gender 6
- 4.2 Age 7
- 4.3 Spending by age and gender 7
- 4.4 Where users search 8
- 4.5 Day of week by device 8

5. Temporal behaviour 9
- 5.1 Sessions by day of week 9
- 5.2 Clicks, bookings and conversion by month 9

6. Conversion analysis 10
- 6.1 The booking funnel 10
- 6.2 What drives conversion 11
- 6.3 Device performance 12
- 6.4 Potential customers lost before booking 12
- 6.5 Illustrative opportunity sizing 13

7. Revenue analysis 13
- Why this matters 14

8. Strategic recommendations 14
- 8.1 Age-targeted, geo-targeted video ads 15
- 8.2 Fix the iPhone booking flow, then scale spend 15
- 8.3 Weekday-focused scheduling 15
- 8.4 Safety and trust features 15
- 8.5 Retargeting and retention programme 15
- 8.6 Improve Web and Moweb experience 16
- 8.7 Suggested 90-day roadmap 16

9. Limitations and next steps 16
- Limitations 16
- Recommended next steps 17

Appendix A. Original dashboard pages 17

Appendix B. Register of approximate values 20

1. Executive summary

This report analyses the four-page Power BI dashboard built for Airbnb's advertisement strategy. Between 5 May 2014 and 23 April 2015, the platform recorded **7,756 visits** from **630 unique visitors**, which produced **984 bookings** and **$245.11K** in revenue, an overall conversion rate of **31.58%**. The analysis covers who the users are, when and how they arrive, where the booking funnel leaks, and which users generate revenue, and it closes with prioritized recommendations for ad spend.

7,756	630	984	$245.11K	31.58%
Total visits	Unique visitors	Total bookings	Total revenue	Conversion rate

Key findings

- **Audience.** The core age group is 25 to 35, peaking near age 30. Ages 22 to 35 spend the most, and spending falls significantly after 40.
- **Gender.** Women are 56.1% of users, but the largest booking totals come from male users.
- **Platform mix.** iPhone (about 3.4K users) and the iOS app (2.21K users) lead traffic, with Desktop second (about 2.0K).
- **Timing.** Monday and Wednesday are the busiest day and weekends run about a quarter lower. Daily sessions peaked in September 2014 and again in December 2014 to January 2015.
- **Conversion.** 59% of searches never reach a message, while 77% of messages become booking requests. iPhone converts at 21% against 47% on Desktop, a 2.2x gap.
- **Revenue.** Returning users generate 84.95% of revenue and new users 15.05%. The average booking is worth $249.10.

Headline recommendations

- Run age-targeted, geo-targeted video ads: premium stays for 30 to 35 and budget stays for 22 to 30.
- Keep investing in iPhone and Desktop traffic, but fix the iPhone booking experience first and improve the interface on other devices.
- Concentrate campaigns on Monday and Wednesday and cut spend on low-activity days such as Saturday and Sunday.
- Add safety and trust features, since women are the majority of users and most searches stop before a message.
- Prioritize retention and retargeting, because returning users drive more than 80% of revenue.

2. Scope, data and method

2.1 The dashboard

The dashboard has four report pages, each with date, device and gender filters that cut across the visuals.

Page	Purpose	Main visuals
EDA	Headline KPIs and the overall traffic picture	KPI cards, date slicer, treemap of users by app, users by device, gender pie, conversion gauge, sessions trend
User Demographics	Who the users are and what they spend	Age distribution histogram, age vs revenue scatter, top 10 streets by searches, day-of-week by device heatmap, average spent
Temporal & Booking Trends	When users engage and what drives conversion	Decomposition tree for conversion, monthly clicks/bookings/conversion line chart, sessions by day of week
Performance & Revenue Insights	Where the funnel leaks and who pays	User conversion funnel, device performance combo chart, potential customers lost by device, revenue share donut

2.2 Data model

The data model contains five tables: **Dates** (calendar, day types and day names), **Locations** (location id, address number, street, city, state, latitude and longitude), **Search Data** (visitor id, session id, session number, user agent, device app combo, Date, minimum timestamp, maximum timestamp, total timestamp, search, message sent, booking request sent, location id, booking amount, left review, next id session, next session number, next user agent, Device App, next date, minimum next timestamp, maximum next timestamp, next search, next message sent, next booking request sent), **Unique Users** (visitor id, name, gender and age), **Stats** (supporting table).

2.3 How the numbers in this report were obtained

The underlying data model is stored in a compressed format, so this report is built from the dashboard as displayed. Every figure falls into one of three groups, and approximate figures are marked with a tilde (~):

Type	Meaning	Examples
Exact	Printed on the dashboard	7,756 visits, 984 bookings, 31.58% conversion, funnel counts, device and browser conversion rates, 84.95% / 15.05% revenue split, $249.10 average spent
Approximate (~)	Read from bar heights or positions in a chart that has no data labels	Users by device, searches and bookings by device, retargeting pool, sessions by weekday (indexed), top 10 street searches
Derived	Calculated here from exact figures	Visits per visitor, revenue per visitor, revenue split in dollars, stage drop-off percentages, opportunity scenarios

2.4 Derived performance metrics

Metric	Calculation	Value
Visits per unique visitor	7,756 / 630	12.3
Bookings per unique visitor	984 / 630	1.56
Revenue per unique visitor	$245,110 / 630	$389
Revenue per visit	$245,110 / 7,756	$31.60
Average revenue per booking	$245,110 / 984	$249.10
Search to message rate	1,279 / 3,116	41.05%
Message to booking-request rate	984 / 1,279	76.93%
Overall conversion	984 / 3,116	31.58%

Each unique visitor visits about 12 times and books about 1.6 times, so repeat behavior is central to this platform. That is consistent with the revenue concentration in returning users shown in section 7.

3. Traffic and platform mix

The EDA page shows where visitors come from. Mobile devices dominate: iPhone is the single largest device with roughly 3.4K users, followed by Desktop at about 2.0K. Adding iPhone, Android phone, iPad and Android tablet gives roughly 4.8K of about 7.1K device-attributed users, so about two thirds of users arrive on a phone or tablet.

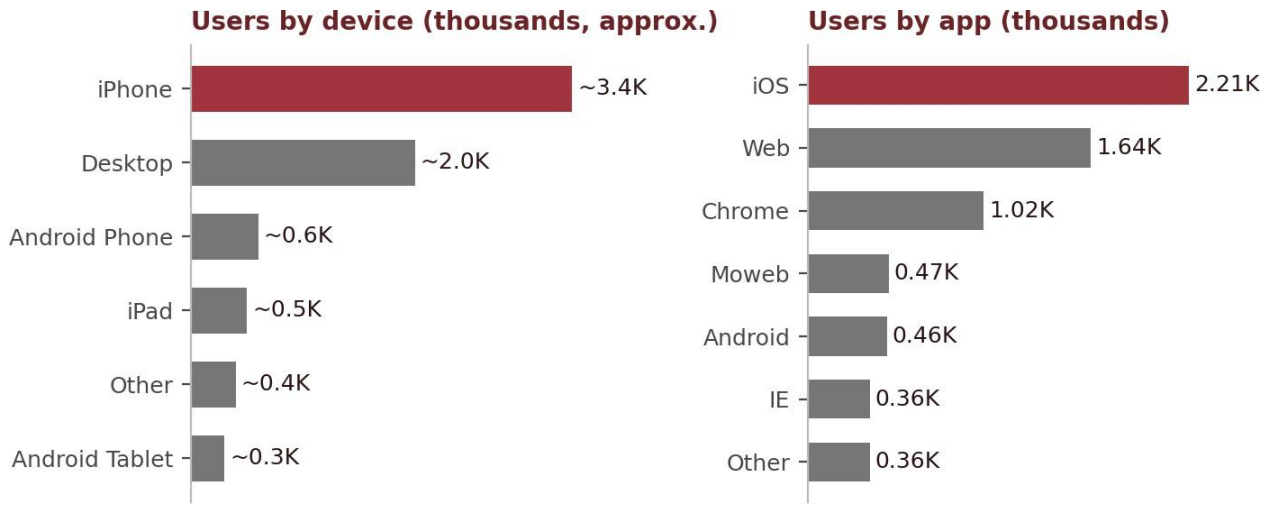


Figure 1. Users by device (approximate) and by app. Highlighted bars mark the leader in each chart.

By app, iOS is the largest segment with 2.21K users, followed by Web (1.64K) and Chrome (1.02K). Moweb (0.47K), Android (0.46K), IE (0.36K) and Other (0.36K) are far smaller. The iOS app brings roughly 4.8 times the users of the Android app, so the audience is heavily skewed to Apple devices.

What this means for advertising

Apple users are the biggest audience and therefore the biggest opportunity and the biggest risk. Ad spend those lands on iPhone traffic only pays off if the iPhone booking experience converts, and section 6 shows that is the weakest part of the funnel.

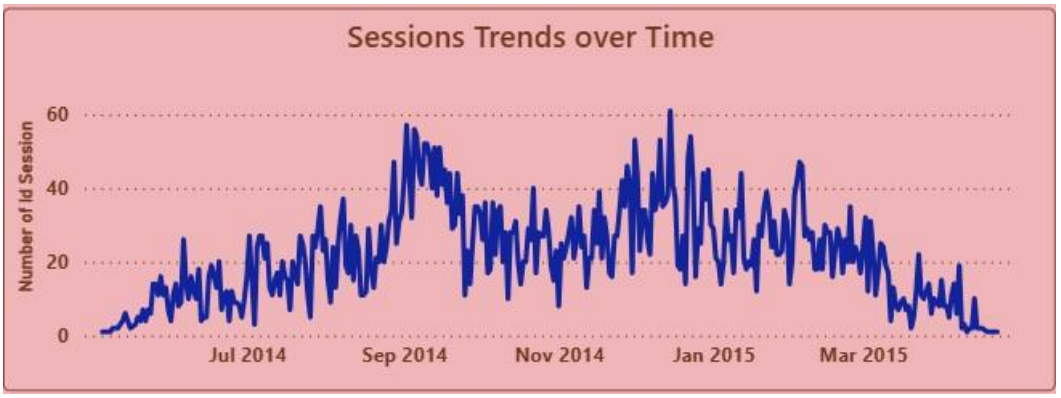
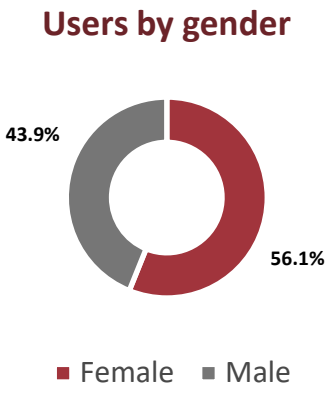


Figure 2. Daily sessions over time.

Daily sessions grew from July 2014, reached about 57 in September, peaked near 60 around December 2014 to January 2015 and then declined into March and April 2015. The dataset spans a single year, so it is not possible to separate a true seasonal cycle from a growth-and-decline pattern.

4. User demographics and spending

4.1 Gender



Women make up **56.1%** of users and men about **43.9%**, so the user base is female-majority. The dashboard's own annotation on the age chart states that male users dominate the platform, but the gender pie and the age histogram both show more women than men. Where men stand out is spending: see section 4.3.

Implication. Safety, trust and communication features matter to the majority of users, and any campaign creative should be tested with female audiences first.

4.2 Age

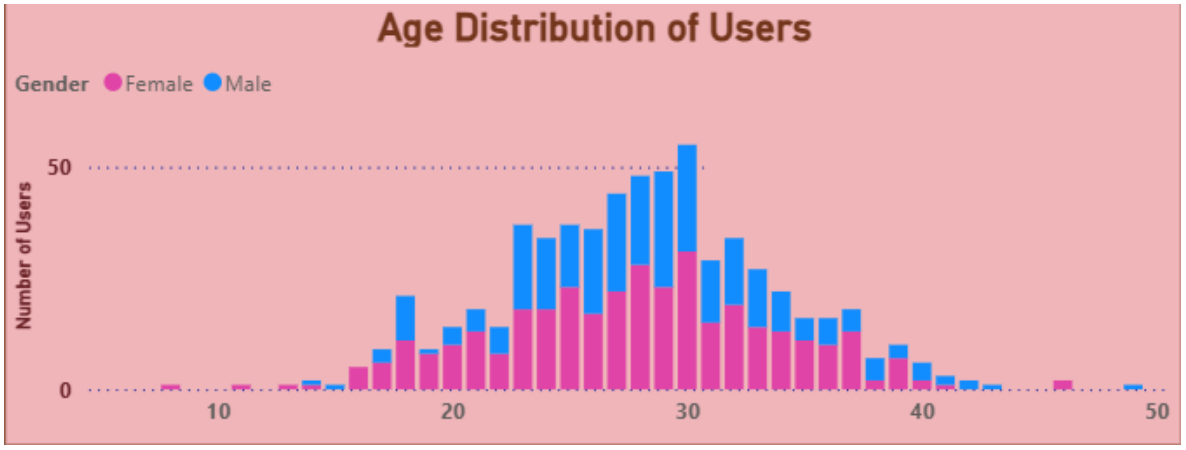


Figure 3. Age distribution of users by gender.

Users are concentrated between 25 and 35 years, with the peak at about 30 (roughly 55 users at that single age). There is a thin tail below 16 and above 45. Women outnumber men at most ages in the core band.

4.3 Spending by age and gender

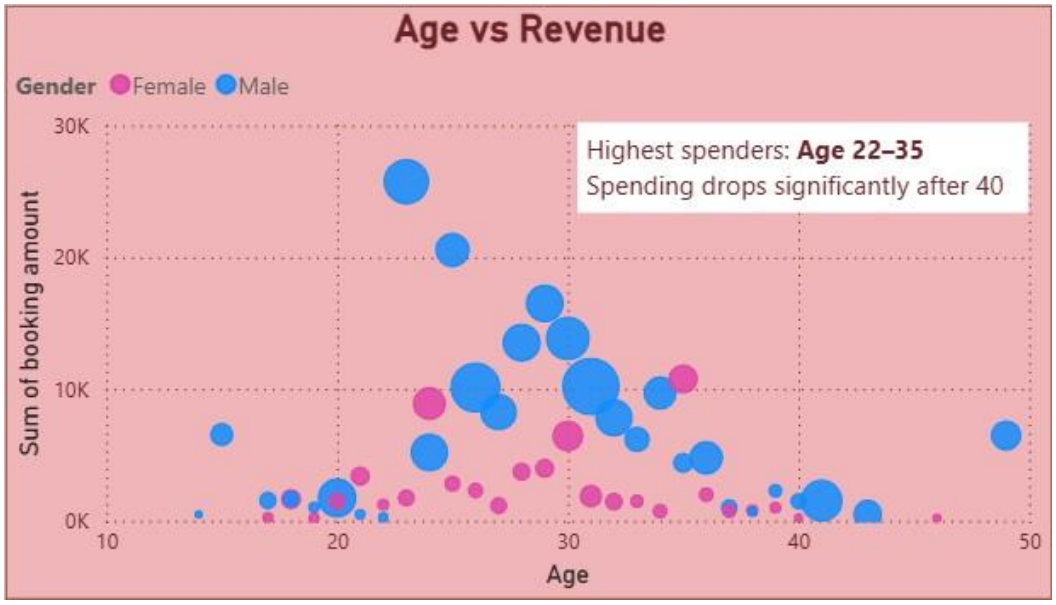


Figure 4. Age vs revenue. Each bubble is one age; bubble size reflects session volume.

- Highest spenders are aged **22 to 35**, and spending drops significantly after 40.
- The largest single-age totals are male users: roughly $26K at about age 23, then about $20K near 25 and $14K to $16K at 28 to 30 (approximate).
- Female totals are lower and peak at roughly $11K near age 35 (approximate).
- The average amount spent per booking is **$249.10**.

4.4 Where users search

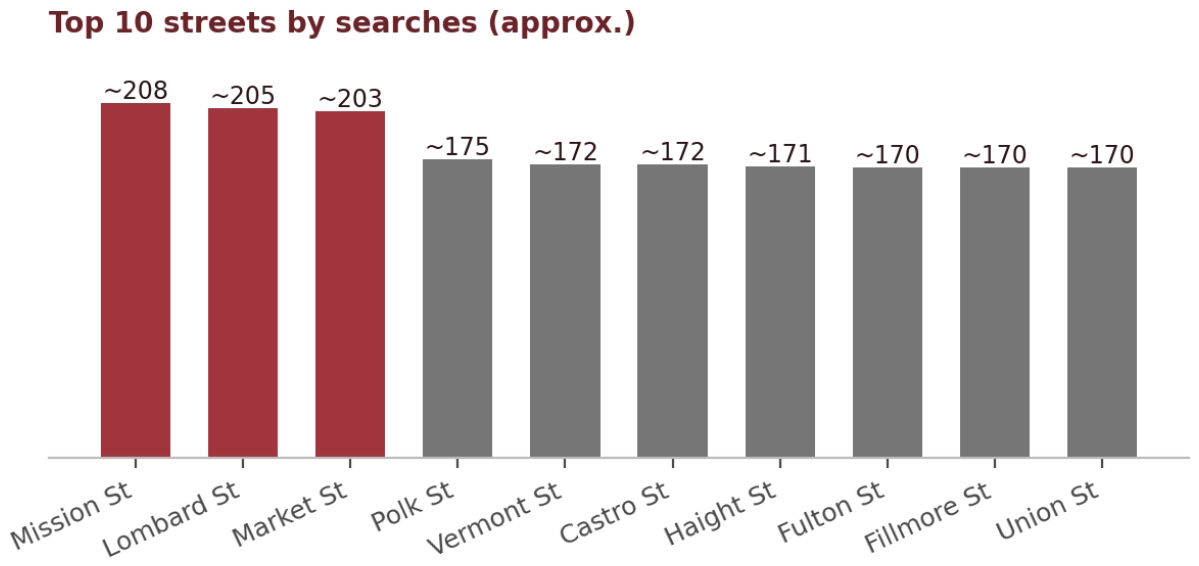


Figure 5. Top 10 streets by number of searches.

Mission, Lombard and Market streets lead with about 200 or more searches each, roughly 15 to 20% ahead of the other seven streets, which sit close together at about 170 to 175. All ten are San Francisco streets, so demand is concentrated in a small set of well-known areas. That supports the geo-targeted approach recommended in section 8.

4.5 Day of week by device

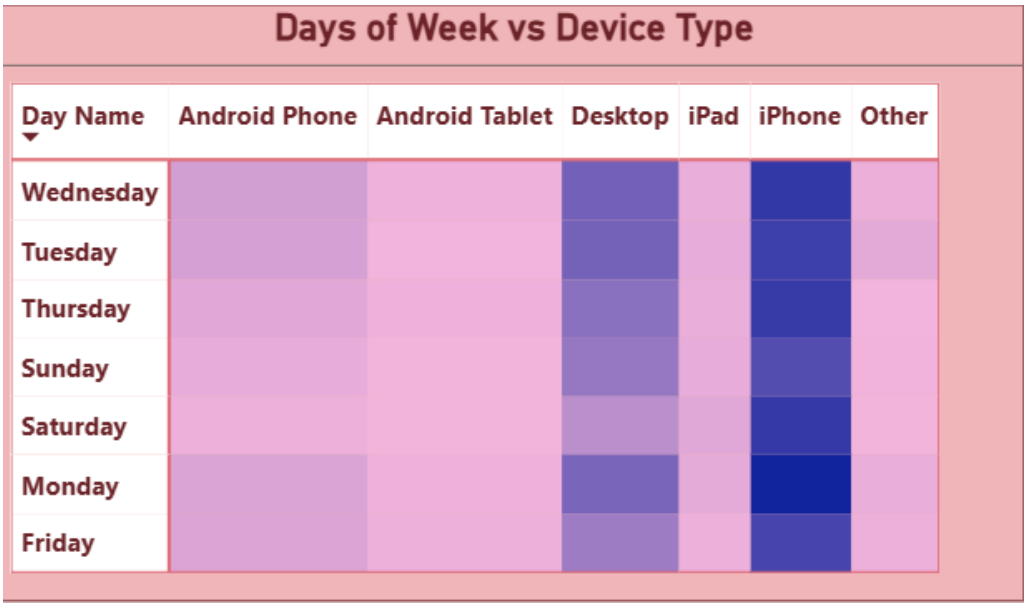


Figure 6. Sessions by day of week and device type.

iPhone is the darkest column on every day of the week and Desktop is consistently second, with the single strongest cell on Monday for iPhone. Android phones, iPads, tablets and other devices are lighter throughout. Device mix therefore does not change much by weekday: the weekday effect is about overall volume, not about a different device audience.

5. Temporal Behavior

5.1 Sessions by day of week

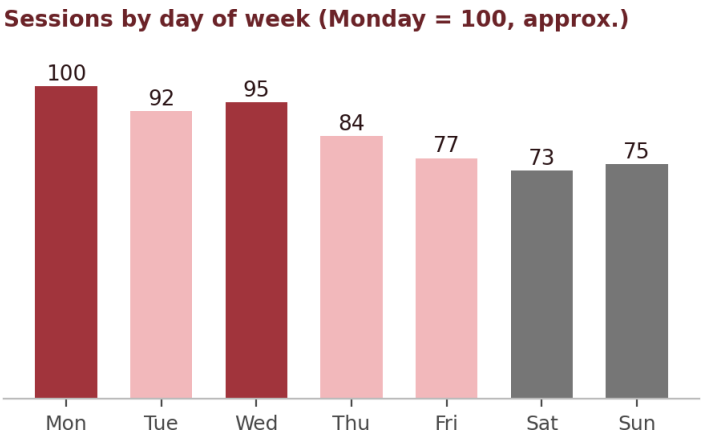


Figure 7. Sessions by day of week, indexed to Monday

Monday is the busiest day, followed by Wednesday and Tuesday. Thursday and Friday are lower, and Saturday and Sunday are the quietest days, running roughly **25% below Monday**.

Bar lengths were converted to an index with Monday = 100 because the dashboard chart has no axis values.

5.2 Clicks, bookings and conversion by month

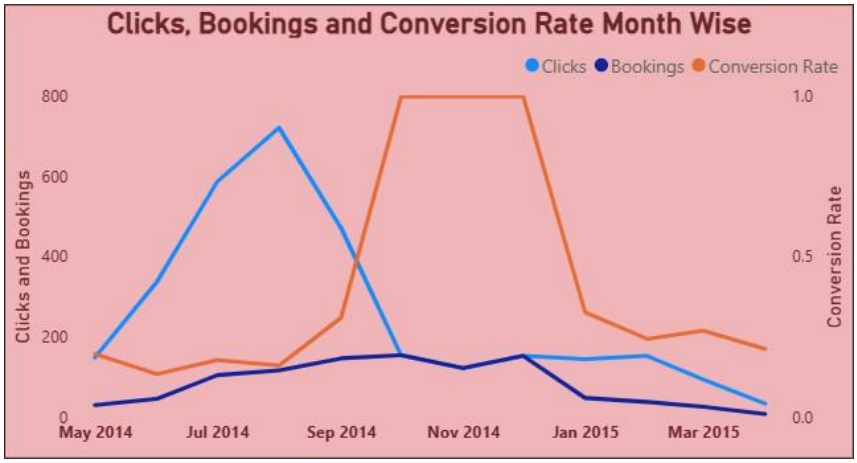


Figure 8. Monthly clicks, bookings and conversion rate.

Period	Clicks (approx.)	Bookings (approx.)	Conversion (approx.)
May to Jun 2014	150 to 340	35 to 50	0.1 to 0.2
Jul to Aug 2014	590 to 720 (peak)	105 to 120	about 0.15
Sep 2014	about 470	about 150	about 0.3
Oct to Dec 2014	about 150	125 to 155	about 1.0
Jan to Apr 2015	150 falling to 35	50 falling to 10	0.2 to 0.3

Bookings stayed in a fairly narrow band of about 100 to 160 per month from July to December 2014 while clicks swung between roughly 150 and 720. The jump in conversion to about 100% in October to December therefore came from clicks falling to the level of bookings, not from more bookings. Conversion measured against clicks can look excellent in a quiet month, so it should always be read next to volume.

Metric definition to confirm

The overall conversion rate on the dashboard (31.58%) equals bookings divided by searches (984 / 3,116). The monthly chart reaches 100%, which suggests it divides bookings by clicks instead. Using one agreed definition in both places would remove a source of confusion.

6. Conversion analysis

6.1 The booking funnel

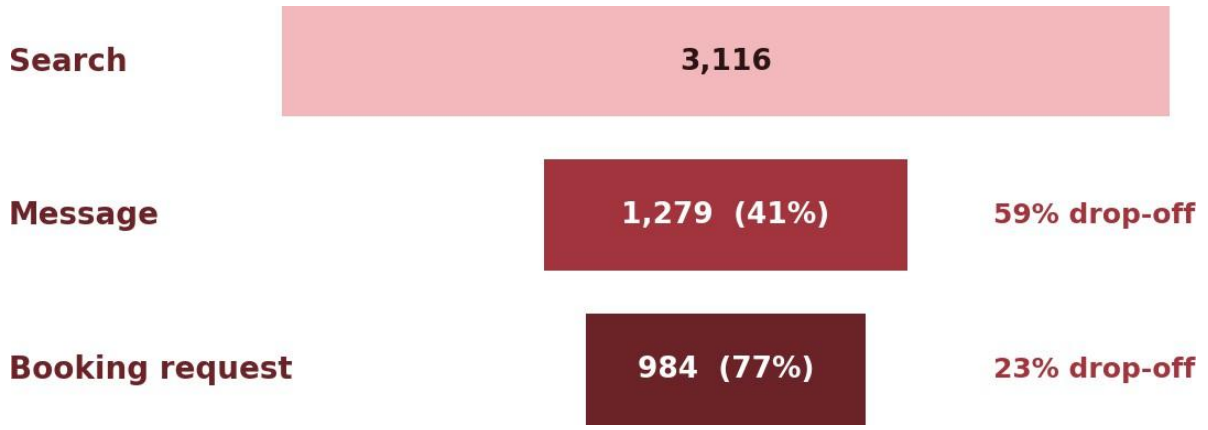


Figure 9. Search to message to booking request.

Of 3,116 searches, 1,279 (41%) become messages and 984 of those messages (77%) become booking requests, giving the overall 31.58%. In absolute terms the largest leak is between search and message, where about 1,837 searches stop. The second stage is healthy by comparison, so the first conversation with a host is the main moment to improve.

6.2 What drives conversion



Figure 10. Decomposition tree: conversion rate (0.32) split by device, then by browser for Desktop.

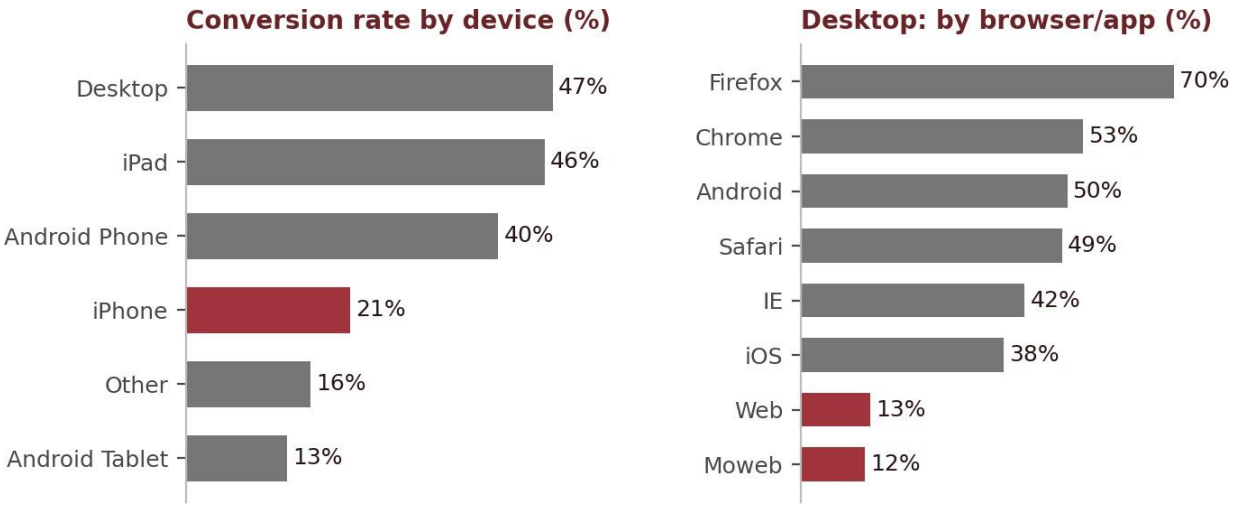


Figure 11. Conversion rate by device, and by browser/app within Desktop.

- Desktop (47%) and iPad (46%) convert best, followed by Android phone (40%).
- iPhone converts at only **21%**, less than half the Desktop rate. Other (16%) and Android tablet (13%) are weaker still. Devices recorded as NULL convert at 31%.
- Within Desktop, Firefox (70%), Chrome (53%), Android (50%) and Safari (49%) are strong. The **Web (13%) and Moweb (12%)** channels are weak spots, even on Desktop.

6.3 Device performance

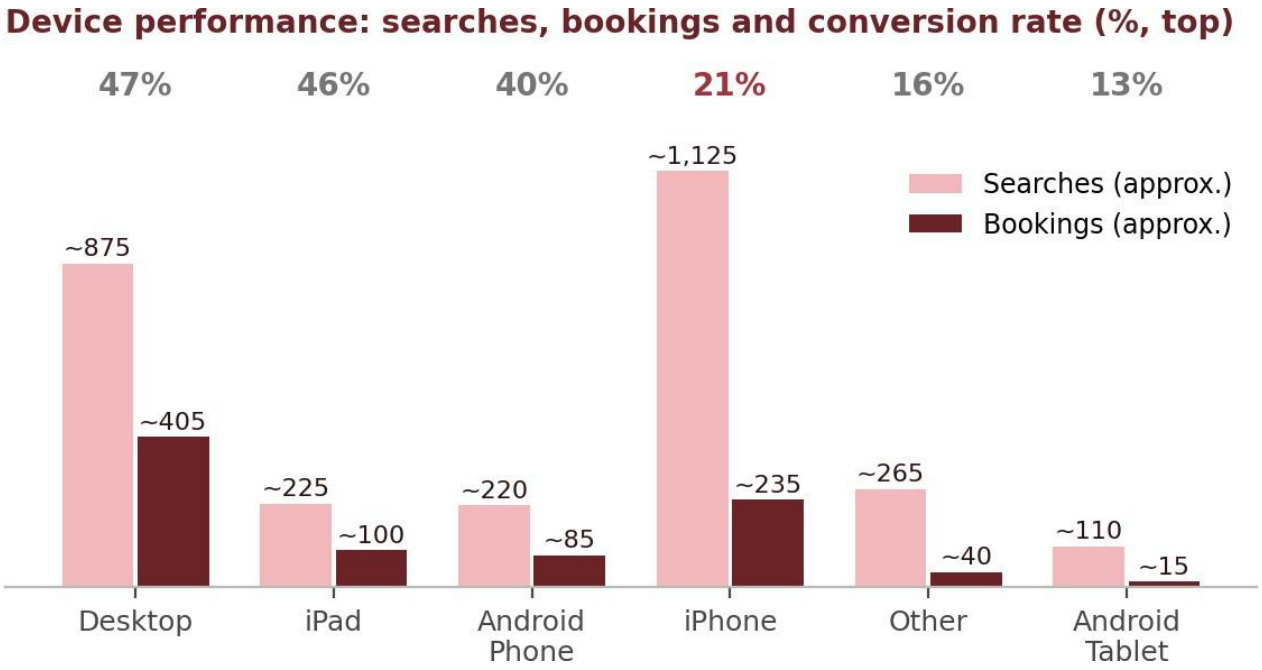


Figure 12. Searches, bookings and conversion by device.

iPhone generates the highest search volume (about 1.1K) but only about 235 bookings. Desktop converts about 405 bookings from fewer searches (about 875). iPhone is the clearest case of high volume with low efficiency.

6.4 Potential customers lost before booking



Figure 13. Retargeting pool by device.

The retargeting pool is the number of searches that did not become a booking, and it matches searches minus bookings for each device. iPhone has about 885, close to half of the roughly 1,920 device-attributed total, and Desktop has about 465. These users already searched, so they are the most reachable audience for remarketing.

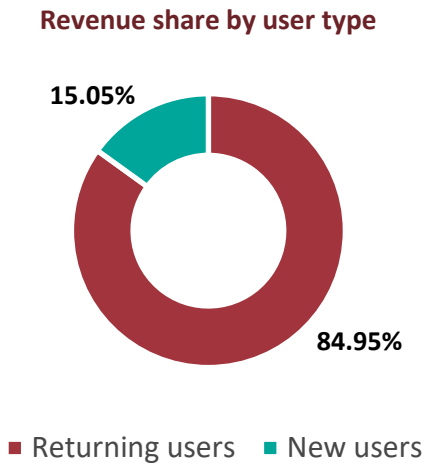
6.5 Illustrative opportunity sizing

The table shows what would happen if iPhone conversion improved toward the rates other devices already achieve. It assumes iPhone search volume of about 1,125 and about 235 current bookings, and uses the average booking value of $249.10 for every device. It is a scenario, not a forecast.

iPhone conversion target	Reference	Extra bookings	Extra revenue	Uplift on total revenue
30%	Midway to Desktop	~102	~$25.5K	~10%
40%	Android phone level	~215	~$53.6K	~22%
47%	Desktop level	~294	~$73.2K	~30%

A smaller lever sits at the top of the funnel: each extra percentage point of search-to-message rate (41% to 42%) would add about 31 messages and, at the current 77% follow-through, about 24 booking requests, worth roughly $6.0K.

7. Revenue analysis



Returning users generate 84.95% of total revenue and new users 15.05%.

Applied to the $245.11K total, that is about **$208.2K** from returning users and about **$36.9K** from new users (derived).

The average booking is worth **$249.10**, which matches total revenue divided by 984 bookings.

Why these matters

- Revenue is highly concentrated: five of every six dollars come from people who have booked before.
- With about 12 visits and 1.6 bookings per unique visitor, loyalty and repeat booking are the engine of the business.
- Acquisition spends aimed at new users has a smaller revenue effect today, so it should be judged on lifetime value, not first-booking revenue.
- The dashboard does not report how many users are returning versus new, so revenue per user type cannot be calculated yet.

Implication for ad strategy

Shift part of the budget toward re-engagement: retargeting the lost-before-booking pool, win-back messages to past guests, and lookalike audiences built from returning users.

8. Strategic recommendations

The recommendations below combine the dashboard's own suggestions with the evidence gathered in this report.

#	Action	Evidence	Metric to track	Priority
1	Age-targeted, geo-targeted video ads	Ages 22 to 35 spend most; 25 to 35 is the core; demand concentrated on a few streets	Conversion and revenue by age band and area	High
2	Fix the iPhone booking flow, then scale spend	iPhone 21% vs Desktop 47%; about 885 lost potential customers	iPhone conversion rate; bookings per 100 searches	High
3	Weekday-focused scheduling	Monday highest; weekends about 25% lower	Cost per booking by weekday	Medium
4	Safety and trust features	Women are 56% of users; 59% of searches never reach a message	Search to message rate; female booking share	High
5	Retargeting and retention programme	Returning users 84.95% of revenue; retargeting pool of roughly 1.9K	Returning-user revenue; repeat booking rate	High
6	Improve Web and Moweb experience	Conversion of only 13% and 12% on those channels	Conversion by browser/app	Medium

8.1 Age-targeted, geo-targeted video ads

Run short video ads on social media aimed at 22 to 35 year olds, with two creative tracks: **premium stays for the 30 to 35 segment** and **budget stays for 22 to 30**. Weight the geography toward the streets with the highest search volume (Mission, Lombard, Market). Spending falls significantly after 40, so older cohorts should receive little or no paid reach.

8.2 Fix the iPhone booking flow, then scale spend

iPhone and Desktop deserve the largest share of ad spend because they carry the most traffic. However, iPhone converts at less than half the Desktop rate, so extra spend there is partly wasted until the experience improves. Audit the iPhone search-to-message and message-to-booking steps, test a simpler booking flow, and improve the interface on other devices (Android phone, tablets and Other). At the Desktop conversion level, iPhone alone could add roughly 290 bookings (about $73K), as sized in section 6.5.

8.3 Weekday-focused scheduling

Concentrate campaigns on Monday and Wednesday, the strongest days, and reduce spend on low-activity days, especially Sunday. Because the device mix barely changes by weekday, the schedule can be applied across devices.

8.4 Safety and trust features

Add visible safety features (verification, secure messaging, host and guest reviews) to bring in more female users and to raise the share of searches that turn into messages. This addresses the largest funnel leak and speaks to the majority of the user base.

8.5 Retargeting and retention programme

Use the potential-customer pool, with about 885 lost iPhone users and 465 on Desktop, as a ready-made remarketing audience. Pair it with win-back campaigns for past guests, since returning users carry more than 80% of revenue.

8.6 Improve Web and Moweb experience

Web and Moweb convert at 13% and 12% on Desktop, against 40% to 70% for the other browsers. Investigate whether this is a technical or a user-experience problem before directing ad traffic to those entry points.

8.7 Suggested 90-day roadmap

Timeframe	Focus	Outputs
Days 1 to 30	Measure and fix	Agree one conversion definition; audit the iPhone flow; set up remarketing audiences; add ad-spend data to the model
Days 31 to 60	Test	Launch age and geo video tests (two creative tracks); weekday scheduling test; trust-feature pilot
Days 61 to 90	Scale	Scale winning creatives and days; extend retargeting; review iPhone conversion against the 30% scenario

9. Limitations and next steps

Limitations

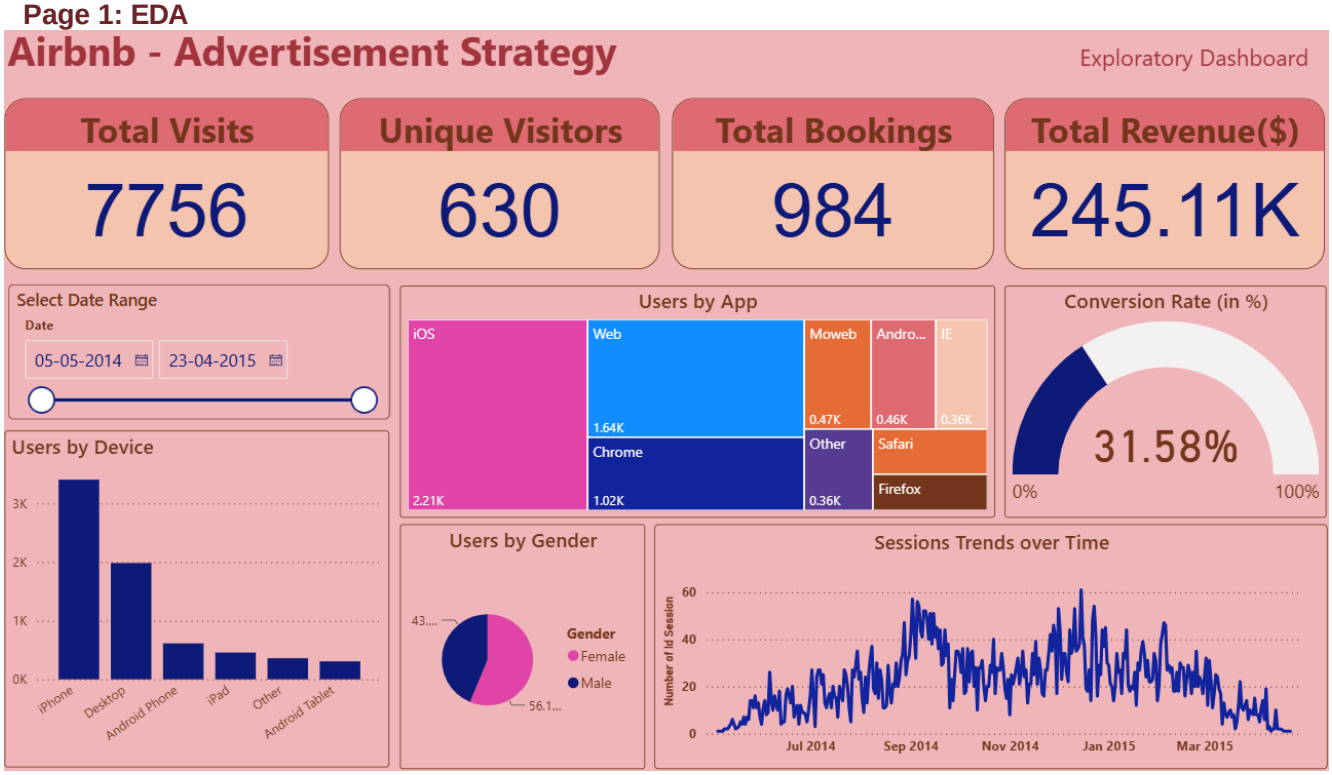
- **Approximate readings.** Several values were read from charts without data labels (marked ~). They are consistent with each other (for example, searches minus bookings reproduces the retargeting pool) but are not exact.
- **One year of data.** The period runs from 5 May 2014 to 23 April 2015, so seasonality cannot be confirmed.
- **No spend data.** The dashboard has no advertising cost or channel data, so return on ad spend cannot be calculated.
- **Small segments.** Some device and browser groups have few sessions, so their conversion rates may move a lot with small changes.
- **Dashboard note vs data.** The note saying male users dominate the platform conflicts with the 56.1% female share shown on the dashboard's own pie chart.
- **Two conversion definitions.** The 31.58% overall rate and the monthly rate reaching 100% appear to use different denominators.

Recommended next steps

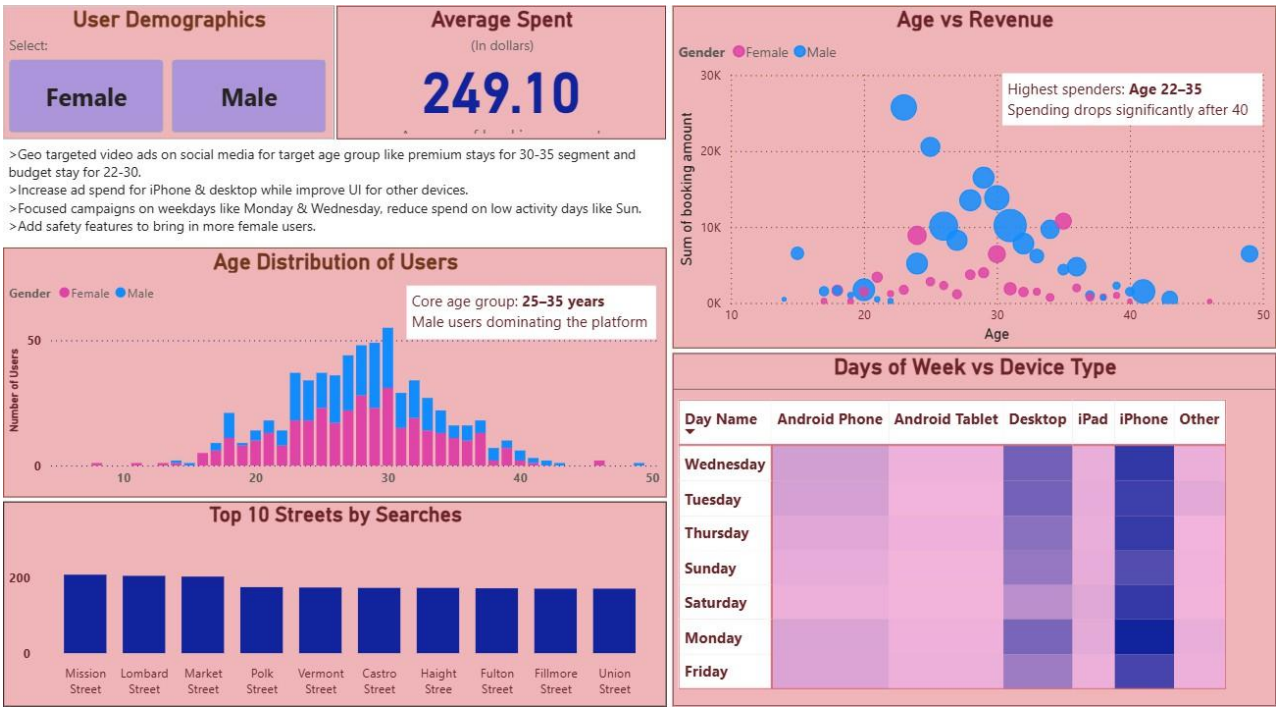
- Add ad cost, campaign and channel fields to the model to measure return on ad spend and cost per booking.
- Add counts of returning versus new users so that revenue per user type and lifetime value can be calculated.
- Add data labels or tooltips to the device, retargeting and weekday charts so that exact values are visible.
- Correct the gender annotation on the age distribution chart and align the conversion definition across pages.
- Extend the dataset beyond one year to confirm seasonality.

Appendix A. Original dashboard pages

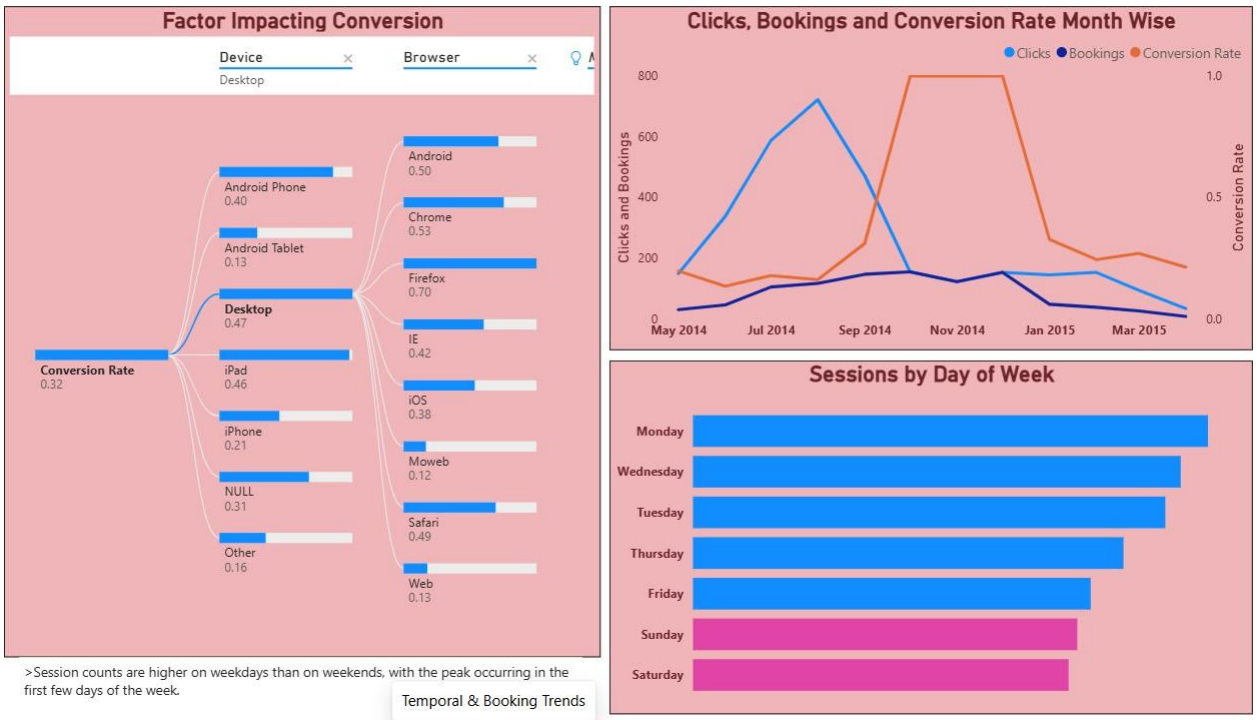
Page 1: EDA



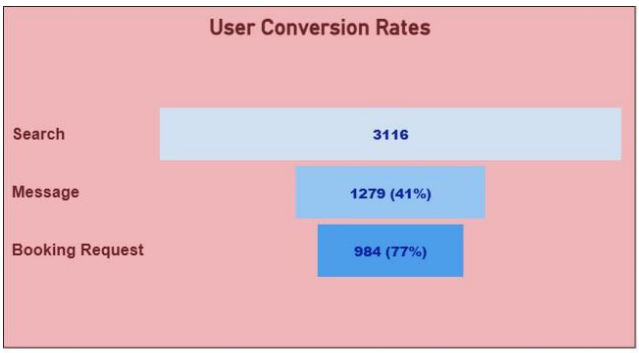
Page 2: User Demographics

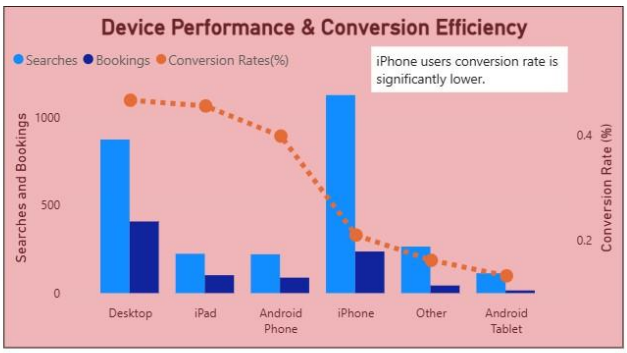


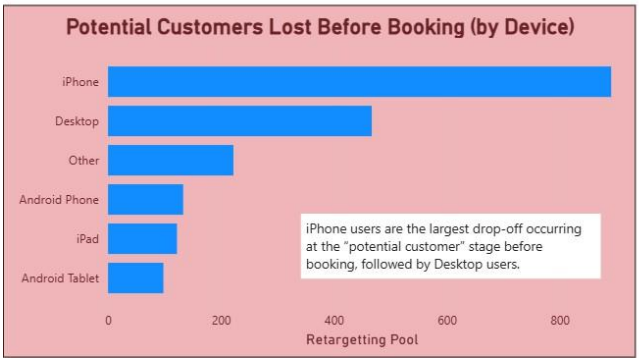
Page 3: Temporal & Booking Trends

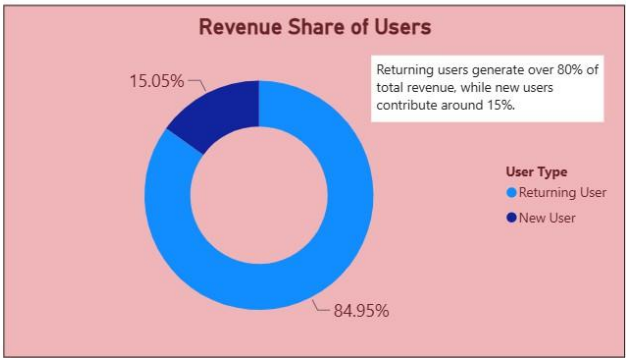


Page 4: Performance & Revenue Insights









Appendix B. Register of approximate values

These figures were read from charts that do not print their values. They are shown rounded and are marked with a tilde in the report.

Chart	Values read
Users by device (thousands)	iPhone ~3.4, Desktop ~2.0, Android Phone ~0.6, iPad ~0.5, Other ~0.4, Android Tablet ~0.3
Searches by device	Desktop ~875, iPad ~225, Android Phone ~220, iPhone ~1,125, Other ~265, Android Tablet ~110
Bookings by device	Desktop ~405, iPad ~100, Android Phone ~85, iPhone ~235, Other ~40, Android Tablet ~15
Retargeting pool	iPhone ~885, Desktop ~465, Other ~220, Android Phone ~130, iPad ~120, Android Tablet ~100
Sessions by weekday (Mon = 100)	Mon 100, Tue 92, Wed 95, Thu 84, Fri 77, Sat 73, Sun 75
Top 10 streets by searches	Mission ~208, Lombard ~205, Market ~203, Polk ~175, Vermont ~172, Castro ~172, Haight ~171, Fulton ~170, Fillmore ~170, Union ~170
Monthly clicks and bookings	Clicks peak ~720 (Aug 2014); bookings ~100 to 160 (Jul to Dec 2014); conversion ~1.0 (Oct to Dec 2014)